

phase_02_GAP_AUDIT

Phase 2 Gap Audit — Console Nodes & Distributed Foundations

Field	Value
Auditor	Engineering auditor (code-grounded)
Date	2026-06-01
Method	Direct inspection of pkg/, internal/, cmd/, api/ + go test of all Phase 2 packages

3-Axis Package Status (Refreshed 2026-06-04, HXC-939)

Why this section exists: The old single-status table below conflated “the package exists” with “the package is used.” A package can be **implemented** and **tested** yet never reached from any binary (**orphaned** / **not wired**). wired here is **measured**, not assumed: `go list -deps ./cmd/... | grep -Fx <module-path>/<pkg> (module = github.com/HelixDevelopment/helix_cluster)`. “**Completed (registry) ≠ wired**” — a package marked Completed in the HXC registry only means source+tests exist; it does NOT prove it is reachable from a shipped binary.

Package	implemented	wired (reachable from cmd/)	tested
pkg/swim	yes	yes	yes
pkg/wireguard	yes	yes	yes
pkg/discovery	yes	yes	yes
pkg/leader	yes	NO (orphaned)	yes
pkg/resources	yes	yes	yes
pkg/scheduler	yes	yes	yes
pkg/session	yes	yes	yes
pkg/session/backends	yes	yes	yes
internal/node	yes	yes	yes

Package	implemented	wired (reachable from cmd/)	tested
internal/console	yes	NO (orphaned)	yes
internal/gpu	yes	yes	yes
internal/scheduler	yes	yes	yes
internal/wireguard	yes	NO (orphaned)	yes
internal/session	yes	yes	yes

Orphan callouts (implemented+tested but NOT reachable from any cmd/binary): pkg/leader, internal/console, internal/wireguard. These pass tests and look “done” in the registry but are not wired into a shipped binary — exactly the exists-but-unused state the old audit hid. Wiring (or an explicit consumer) is required before they count as end-user-usable per CLAUDE-1.

Verdict: ~80% complete

Summary: All P0/P1 distributed-foundation packages (swim, wireguard, discovery, leader, resources, scheduler, session, node agent) are genuinely implemented with real-behavior tests that all pass; the remaining ~20% gap is concentrated in console boot coordination (no `linux_boot.go`), real NAT traversal (STUN/UPnP/hole-punching is a stub), and live-migration strategies (CRIU/DMTCP/container correctly deferred). No PASS-bluffs found in the implemented core — tests assert sink-side behavior, not just non-panic.

All 14 Phase 2 packages compiled and passed `go test` at audit time (swim 3.2s, wireguard 0.7s, discovery 1.6s, leader 1.1s, resources 1.3s, scheduler 1.0s, session 1.7s, session/backends 3.7s, internal/node 2.3s, internal/console 2.0s, internal/gpu 3.1s, internal/scheduler 3.0s, internal/wireguard 2.6s, internal/session 2.6s).

Deliverable Status

Deliverable	Status	Evidence (file:line)	Notes
pkg/swim gossip: membership, suspicion, indirect ping	DONE	pkg/swim/ suspicion.go:99 (SuspectWithRelays), pkg/ swim/prober.go:55 (IndirectProbe), pkg/swim/ failure_detector.go:38; 3136 LOC	Real incarnation/refute logic; suspicion_test + package_test assert Suspect/Confirm/Refute transitions.
pkg/wireguard mesh: tunnels, key exchange, config gen	DONE	pkg/wireguard/ config.go:37 (GenerateKeyPair via wgtypes/curve25519), pkg/ wireguard/mesh.go, pkg/ wireguard/configgen.go; 1807 LOC	Real key gen + config rendering; mesh+manager tested.
WireGuard key rotation	DONE	pkg/wireguard/ keyrotation.go:104	Generation tracking + supersede ordering;

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		(RotateKeysTracked), :17 (KeyFingerprint), :44 (Supersedes)	keyrotation_config_test covers it.
WireGuard NAT traversal (STUN/UPnP/hole-punch)	PARTIAL	pkg/wireguard/nat_traversal.go:18 ("STUN-like"), :33 & :41 return "UPnP/NAT-PMP not implemented"	DiscoverExternalAddress hits external HTTP echo only; no real STUN client, UPnP, or UDP hole-punching. Roadmap P2 risk item.
pkg/discovery registry + health-aware routing	DONE	pkg/discovery/selection.go:15 (WeightedSelector), :110–114 (healthy + TTL-expiry filter), pkg/discovery/etcd_backend.go; 1485 LOC	In-memory + etcd backends real; weighted health-aware selection tested. etcd_backend_test uses a mock client (real etcd proven in pkg/etcd integration tests per repo).
pkg/leader TTL election + fencing	DONE	pkg/leader/fencing.go:117 (AcquireLease), :45 (IsStale), :73 (LeaseRegistry.acquire); 796 LOC	Monotonic fencing tokens + lease TTL; fencing_test asserts stale-token rejection.
pkg/resources aggregator (CPU/mem/GPU/cgroup)	DONE	pkg/resources/proc_linux.go:40 (/proc/cpuinfo), :93 (/proc/meminfo), pkg/resources/cgroup_v2.go; 1449 LOC	Real /proc + cgroup v2 parsing; proc_mock enables fixture-based tests. GPU is a struct field populated by console layer (see below).
pkg/scheduler plugin framework + optimistic concurrency	DONE	pkg/scheduler/plugins.go:13 (Filter/Score/Bind), pkg/scheduler/scheduler.go:154 (ScheduleOptimistic + version), pkg/scheduler/cost_gpu.go, gang_preempt.go; 2093 LOC	NodeResourcesFit/ CapabilityMatch/ LoadAware plugins + cost-aware GPU + gang/ preempt; version CAS tested.
pkg/session CRDT state + manager	DONE	pkg/session/crdt.go, pkg/session/convergence.go, pkg/session/lifecycle.go; 3058 LOC	CRDT convergence + lifecycle tested with real merge assertions.
pkg/session/backends tmux + native PTY	DONE	pkg/session/backends/native.go:15 (creack/pty), :53 (pty.Start), pkg/session/backends/	Native PTY spawns real process; tmux shells out. tmux Attach is a documented partial (window-pipe, not full

Deliverable	Status	Evidence (file:line)	Notes
Session migration (CRDT / CRIU / DMTCP / container)	PARTIAL	tmux.go:20 (LookPath tmux) pkg/session/migration.go:51 (crdtStrategy real), :178/:196/:214 return "... not implemented"	PTY wrap) — tmux.go:67–73. Only CRDT (state-only) strategy is real; CRIU/DMTCP/container are explicit stubs — matches roadmap §6 P2 “CRIU/DMTCP research, Phase 3”.
internal/node agent: heartbeat, probes, SWIM+WG wiring	DONE	internal/node/node.go:84 (swim.NewProtocol), :97 (wireguard.NewManager), :100 (resources aggregator); node_behavior_test + server_behavior_test	Genuinely composes the three substrates; behavior tests present.
internal/console PS4/PS5 detection + thermal + trust	DONE	internal/console/detector.go:46 (Detect, /proc/cpuinfo + device-tree), thermal.go, trust.go, register.go	Real /proc parsing with injectable paths; detector_test/thermal_test/trust_test/register_sink_test (sink-side) all present.
internal/console GPU compute wrapper (gpu_wrapper.go)	DONE (renamed)	internal/console/gpu.go:21 (GPUQuerier, vulkanLoaderPath); gpu_test + gpu_helpers_test	Vulkan-loader probe; satisfies roadmap gpu_wrapper.go mapping under a different filename.
internal/console Linux boot coordination (linux_boot.go)	MISSING	No linux_boot.go / LinuxBoot symbol in internal/console/	Roadmap §4.3 maps console_dim04 → linux_boot.go. No boot coordination code exists.
internal/gpu manager/monitor	DONE	internal/gpu/manager.go, monitor.go; manager_test + manager_extra_test api/v1/	Generic GPU manager beyond console; tested.
Proto APIs (node/session/scheduler/health)	DONE	node.proto+node.pb.go, session, scheduler, health, advisory, security, build (+_grpc.pb.go)	Generated stubs present and consumed by internal/* servers.
cmd/helix-node, helix-scheduler, helix-session binaries	DONE	cmd/helix-node, cmd/helix-scheduler, cmd/helix-session present (all cmd/* hardened)	Binaries wired to internal servers.
PS4/PS5 bare-metal Linux agent	DEFERRED	n/a	Requires owned jailbroken hardware + firmware; roadmap §7

Deliverable	Status	Evidence (file:line)	Notes
(real hardware join)			high-risk. Detection logic exists but real-node E2E un-runnable here.
Console GPU as CUDA/ROCm compute donor	DEFERRED	n/a	Needs GPU kernels + drivers on console hardware.

Top Implementable Gaps (Go, no new infra)

1. **internal/console/linux_boot.go — boot coordination state machine.** Implement a BootCoordinator that reads kernel/firmware markers (e.g. /proc/version, /sys/firmware/devicetree, cmdline) via injectable paths and reports a BootState (Unknown/Booting/UserspaceReady/ClusterReady) plus a WaitReady(ctx) that gates node registration until probes pass. Pure file-reads + a deterministic state machine; no hardware needed for the logic. *Acceptance test (mutation-pairable):* feed fixture path-sets representing each boot phase and assert the returned BootState; mutant = flip the readiness threshold/ordering and confirm the test fails.
2. **pkg/wireguard real STUN client in nat_traversal.go.** Replace the HTTP-echo DiscoverExternalAddress with a minimal RFC 5389 STUN Binding request over UDP (parse XOR-MAPPED-ADDRESS) against a configurable STUN server; keep UPnP returning a typed ErrUnsupported. No new infra — uses a stock public STUN endpoint or a local test server. *Acceptance test:* spin an in-process fake STUN UDP responder returning a known XOR-mapped addr; assert the parsed reflexive ip:port. Mutant = corrupt the XOR-decode mask and confirm failure.
3. **pkg/discovery etcd-backed integration test (real server).** Add a build-tagged integration test that runs EtcdBackend against a real embedded/containerized etcd (repo already proves etcd in pkg/etcd), asserting register→lookup→TTL-expiry→deregister round-trips — closing the mock-only gap on the backend. *Acceptance test:* register an instance with short TTL, sleep past TTL, assert Lookup excludes it; mutant = drop the lease/TTL on Put and confirm the expired instance wrongly persists.
4. **pkg/session/migration.go container-checkpoint strategy (CRDT+manifest, no CRIU).** Implement containerStrategy to do a state-only checkpoint: serialize CRDT session state + a restart manifest (cmd, env, cwd) to a blob, “migrate” by transferring the blob and re-spawning via the native backend on target. This is real and infra-free (process restart, not live freeze), distinct from the deferred CRIU path. *Acceptance test:* checkpoint a session, restore on a second manager instance, assert CRDT state + manifest equality and a live re-spawned PTY. Mutant = omit env from the manifest and confirm the restored-state assertion fails.
5. **pkg/resources Linux GPU reader (/sys + NVML-free sysfs).** Add a GPUReader that enumerates /dev/dri/card* and /sys/class/drm/*/device (vendor/device IDs, VRAM where exposed) to populate GPUInfo.Count/vendor without vendor SDKs; wire into NodeAggregator (currently GPU is only copied through at aggregator.go:122). *Acceptance*

test: point the reader at fixture sysfs trees (NVIDIA, AMD, none) and assert detected count/vendor. Mutant = swap the vendor-ID map and confirm misclassification fails the test.

6. **pkg/wireguard key-rotation grace/overlap window in mesh.** Extend RotateKeysTracked to keep the superseded key valid for a configurable overlap so in-flight peers don't drop; expose ActiveKeys() returning current+grace keys consumed by mesh.go peer config. *Acceptance test*: rotate, assert both keys are accepted within the window and the old key is rejected after it (driven by an injected clock). Mutant = expire the old key immediately and confirm the overlap assertion fails.

Deferred (need external infra / hardware / Zig / GPU kernels)

- **PS4/PS5 bare-metal Linux agent join (real hardware)**: needs owned, jailbroken consoles + tested firmware; detection logic is implemented but a real-node E2E cannot run in this environment. (Roadmap §7 high-risk.)
- **Console GPU CUDA/ROCm compute donor path**: requires GPU kernels (C/C++) and console GPU drivers; out of scope for Go-only work.
- **CRIU/DMTCP live process migration**: requires CRIU/DMTCP system binaries + privileged checkpoint/restore; roadmap §6 explicitly schedules this for Phase 3 research.
- **Real WireGuard kernel-module mesh over NAT across real networks**: STUN client (gap #2) is implementable, but cross-NAT hole-punching validation needs multi-host/relay infra.

Deferred count: 4 (all blocked on owned console hardware, GPU kernels/drivers, privileged checkpoint binaries, or multi-host networks — none implementable as pure no-infra Go here).